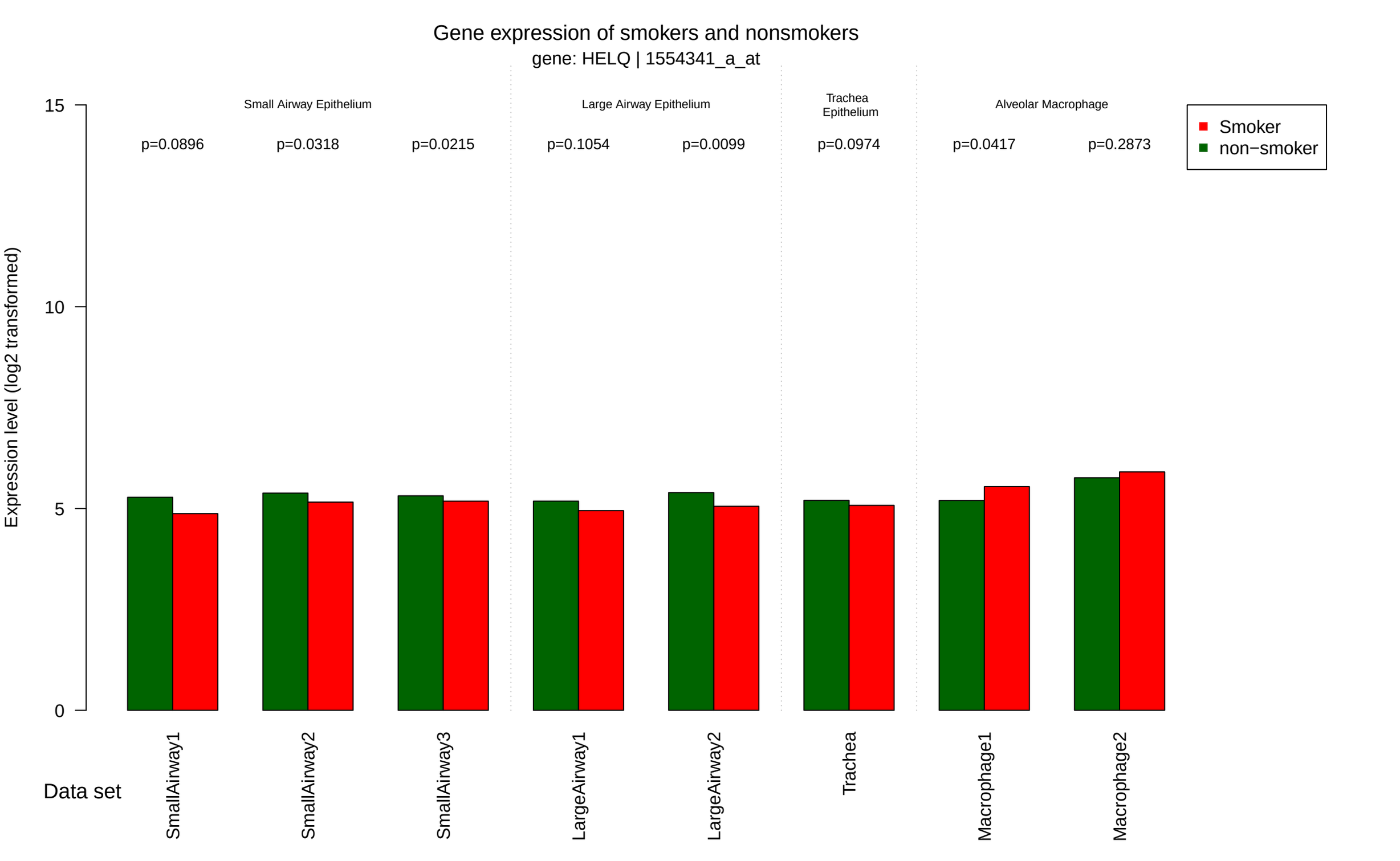
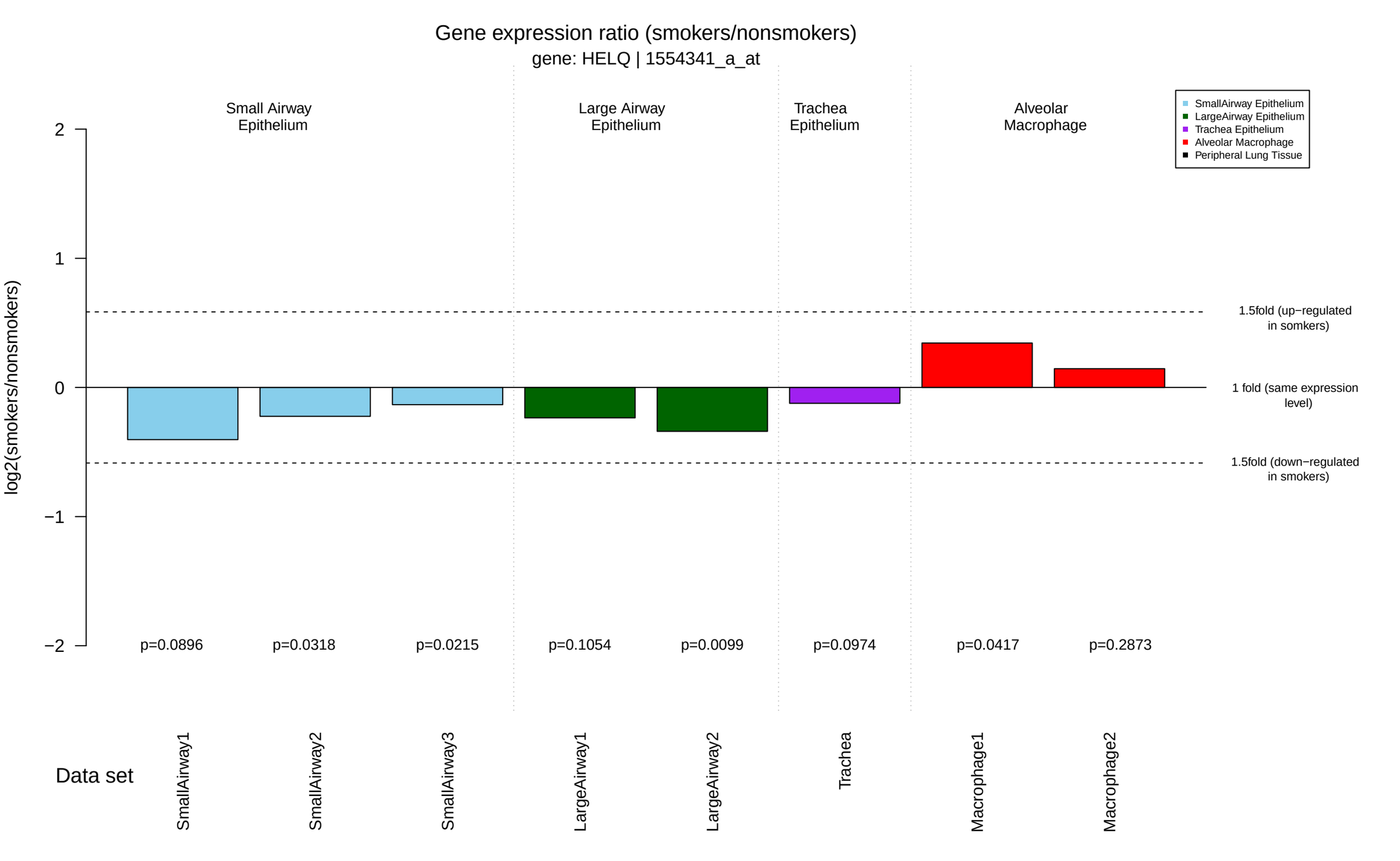
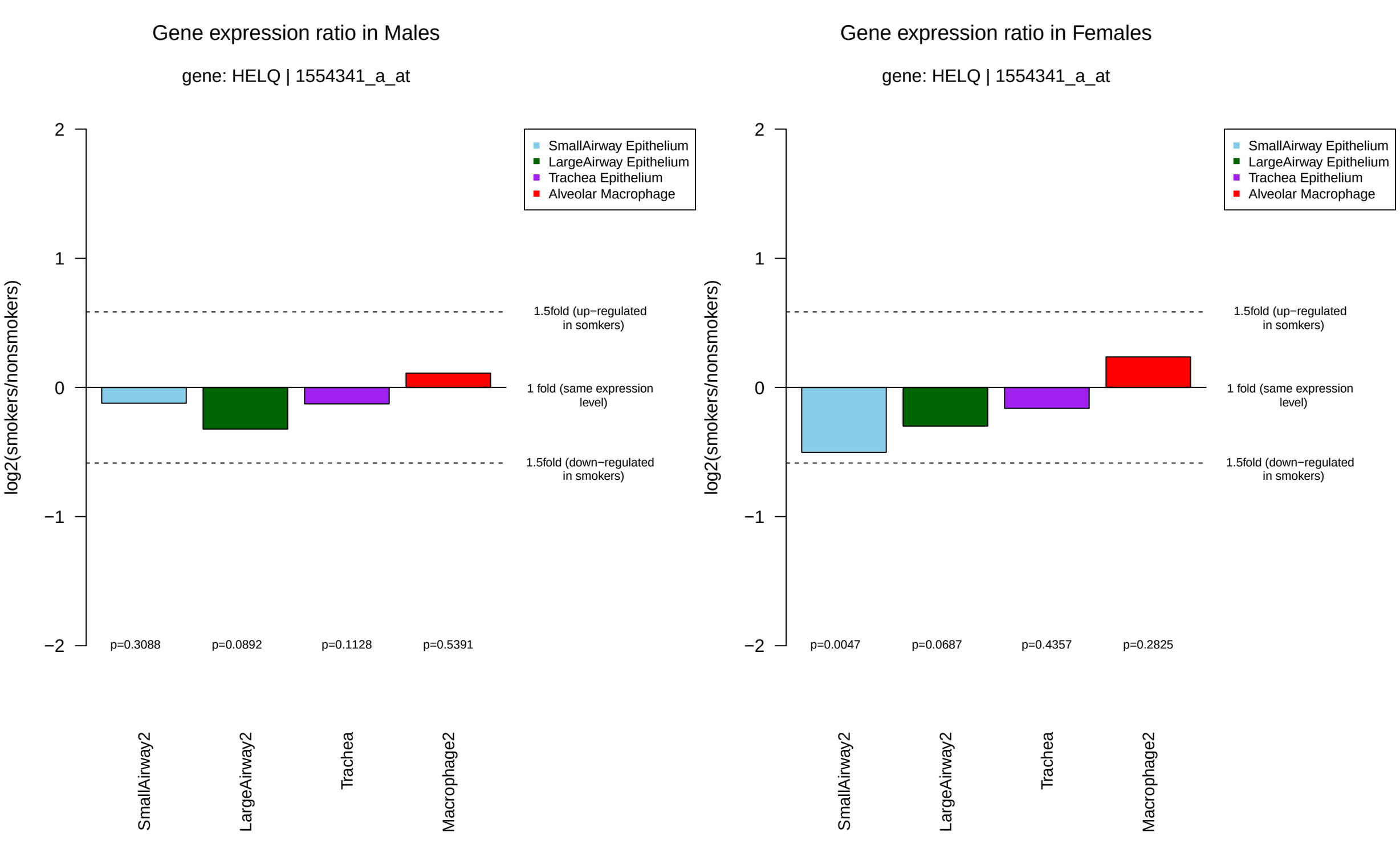
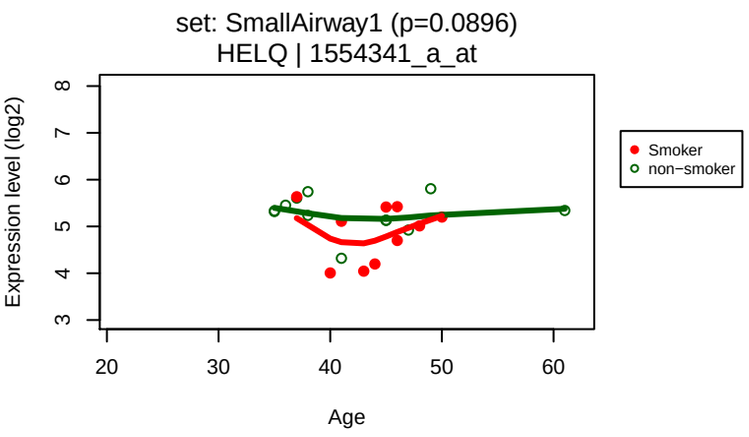
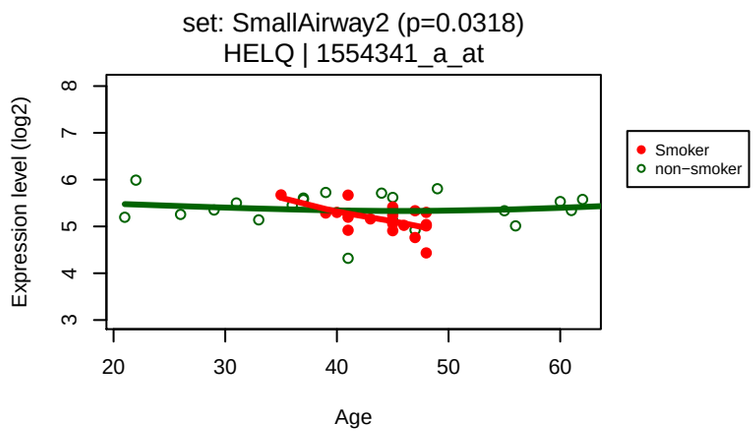
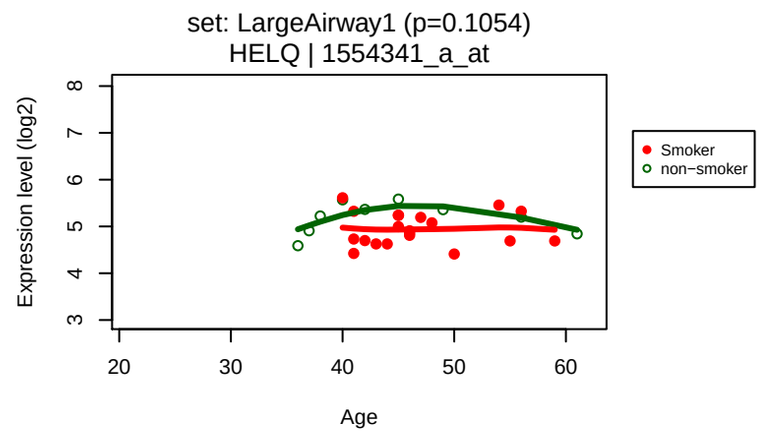

Gene expression of smokers and nonsmokers
gene: HELQ | 1554341_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0896
p=0.0318
p=0.0215
p=0.1054
p=0.0099
p=0.0974
p=0.0417
p=0.2873
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

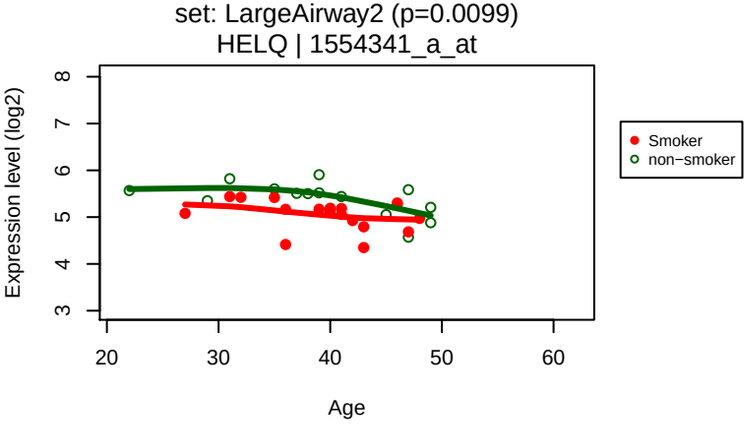

Gene expression ratio (smokers/nonsmokers)
gene: HELQ | 1554341_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0896
p=0.0318
p=0.0215
p=0.1054
p=0.0099
p=0.0974
p=0.0417
p=0.2873
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

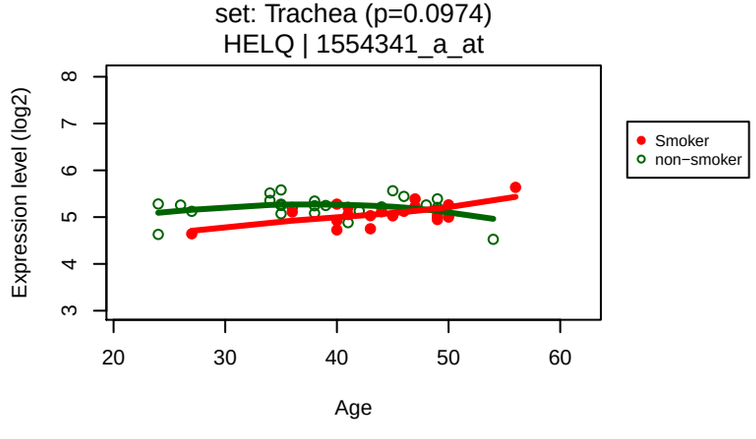

Gene expression ratio in Males
gene: HELQ | 1554341_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3088
p=0.0892
p=0.1128
p=0.5391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HELQ | 1554341_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0047
p=0.0687
p=0.4357
p=0.2825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

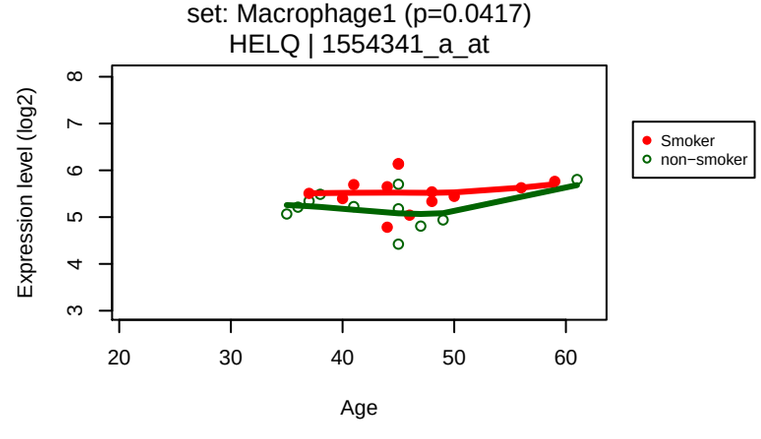

set: SmallAirway1 (p=0.0896)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker

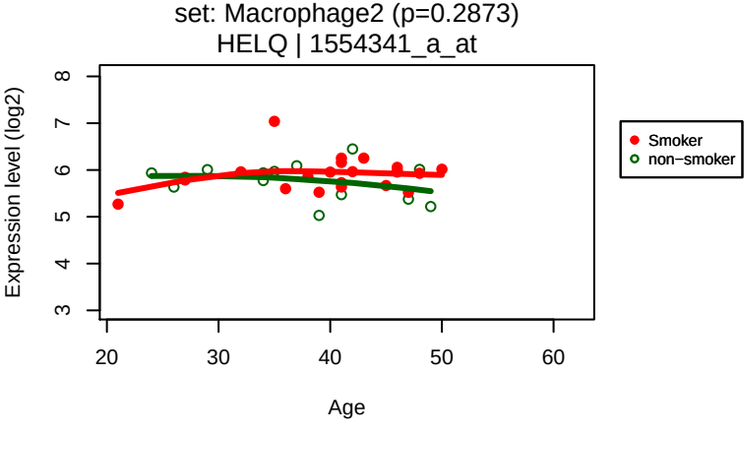

set: SmallAirway2 (p=0.0318)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker

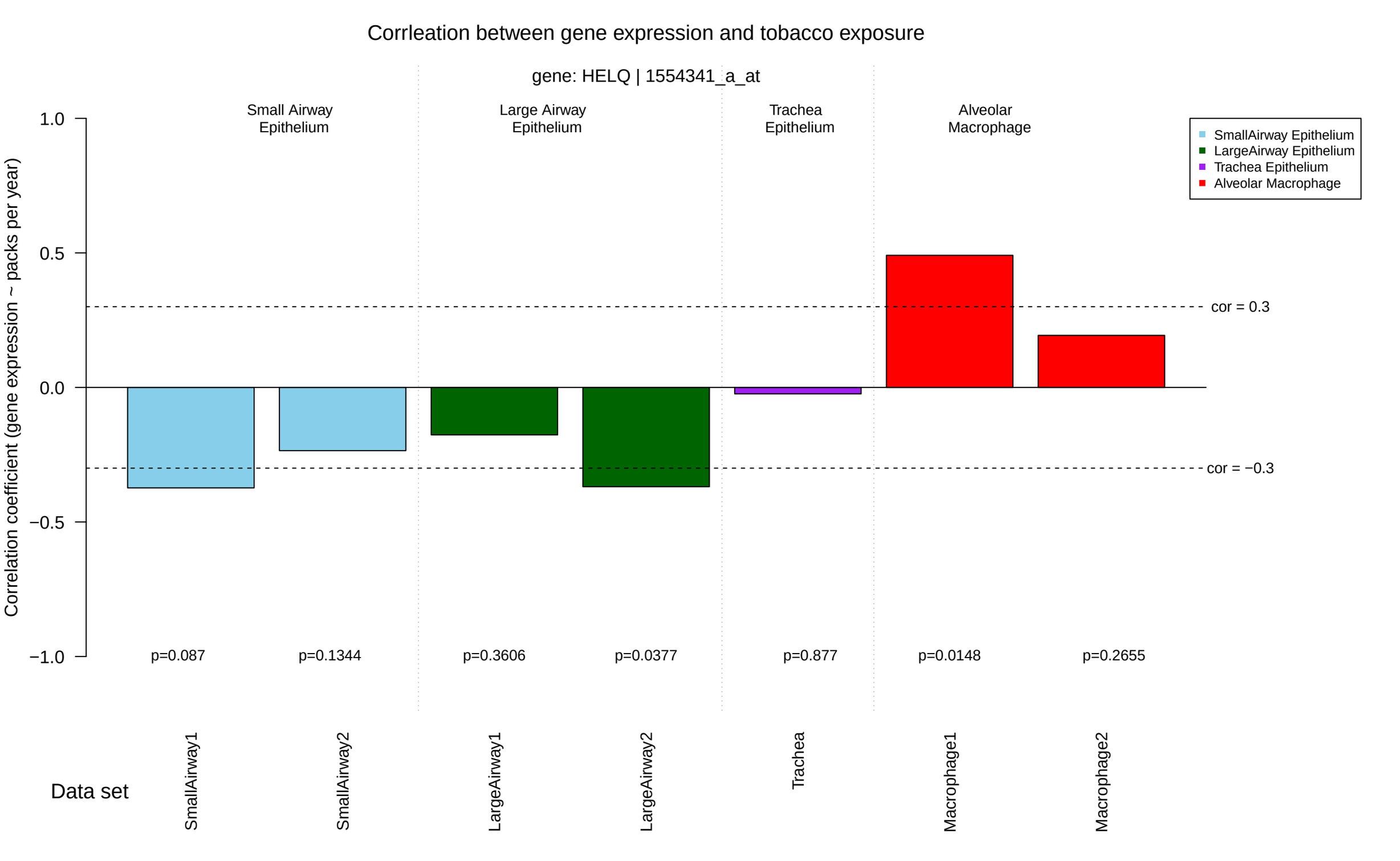

set: LargeAirway1 (p=0.1054)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0099)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0974)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0417)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2873)
HELQ | 1554341_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HELQ | 1554341_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.087
p=0.1344
p=0.3606
p=0.0377
p=0.877
p=0.0148
p=0.2655
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2